

Do Register Tokens Regularize Vision Transformers Under Data Scarcity?

A controlled 12-run ablation on low-data CIFAR-100

Shahin Gulnisa Narmina Emil Rufet

01

The question

Shahin | Architecture & training lead

Large ViTs quietly vandalise their own attention

Darcet et al. (ICLR 2024) found that in large vision transformers a handful of patch tokens acquire enormous activation norms. The [CLS] token routes much of its attention to them.

Those tokens sit in *empty* image regions and carry global information. The model is using the image as scratch paper.

Their fix is architectural and almost free: give the model dedicated scratch tokens — *registers* — and the artifacts disappear.

The gap in that story

It was shown where data is *abundant*. If registers absorb capacity the model would otherwise spend memorising, they should behave like a **regularizer** — and a regularizer only shows itself where overfitting bites.

So: when a ViT is starved of data, do registers reduce the generalization gap?

Three hypotheses, each falsifiable

H1

Structural regularization

Registers shrink the gap

$$\Delta\mathcal{L} = \mathcal{L}_{\text{val}} - \mathcal{L}_{\text{train}}.$$

H2

Capacity dilution

At $d = 192$ a large K eats representational room, so accuracy turns non-monotonic in K .

H3

Entropy collapse

Registers prevent the layer-wise collapse of attention entropy that accompanies artifact formation.

Why this is worth an experiment

Nobody has tested registers as a regularizer, at small scale, under a strict data budget. A clean answer is useful whichever way it falls.

02

Starving the model

Gulnisa | Data pipeline

A deliberately small slice of CIFAR-100

100 images per class, stratified, drawn from the 50,000-image train split.

Training	9,000 images
Validation	1,000 images
Test, untouched	10,000 images

Upsampled to 224×224 so the pretrained patch grid applies unchanged. RandomResizedCrop, horizontal flip, AutoAugment; evaluation sees resize and centre crop only.

The design decision that makes it a controlled test

The subset is drawn *under the run's own seed*. So a seed is a replicate of the **whole pipeline**, not just of initialisation — and within a seed, all four register arms see the **identical** images in the identical order.

One thing moves, and only one

Held fixed across every arm:

- the same 9,000 training images
- the same train/validation split
- the same batch order
- the same 50-epoch schedule, optimiser and augmentation

Varied: K , **the number of register tokens**.
Nothing else.

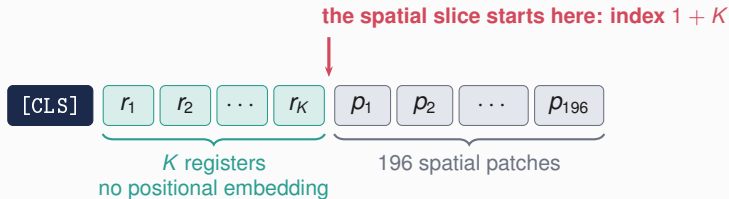
Why that matters at $n = 3$

Between-seed spread in validation accuracy is **2.17 pp** — far larger than any effect registers could plausibly have. Because the arms are matched within a seed, we can subtract that noise away instead of drowning in it.

03

What a register is, and how we measure it

The token sequence, and the slice that decides everything



Registers join every self-attention block and are discarded before the head. Cost at $K = 8$: **1,536 of 5.54 M** parameters — a relative increase of 2.8×10^{-4} .

The off-by- K trap

Slice the spatial row from index 1 instead of $1 + K$ and the register columns leak into the map. It still looks plausible — which is exactly why every metric asserts the slice width.

Two mechanism readouts

Layer-wise attention entropy

$$\bar{H}^{(l)} = -\frac{1}{BHS} \sum_{b,h,i} \sum_j A_{b,h,i,j}^{(l)} \log_2 A_{b,h,i,j}^{(l)}$$

Attention rows are probability distributions, so their entropy measures how concentrated the routing is. Artifacts should show up as a *fall* in the deep blocks.

Modern `timm` runs fused attention and never materialises the $[B, H, S, S]$ matrix — our hook reconstructs it from the block's own Q and K projections.

Patch-norm outlier rate

$$\rho^{(l)} = \frac{1}{BN} \sum_{b,n} \mathbf{1} \left[\|h_{b,K+n}^{(l)}\|_2 > \mu_b^{(l)} + 3\sigma_b^{(l)} \right]$$

This is the artifact definition itself: how many patch tokens carry an anomalously large norm.

Both are read from the **same 64 test images** in all twelve runs, so every comparison is made on identical inputs.

04

Twelve runs, no surprises

Emil | Sweep execution & operations

The ablation matrix

Seed	Register count K			
	0	1	4	8
42	■	■	■	■
1337	■	■	■	■
3407	■	■	■	■

12 of 12 completed

ViT-Tiny, ImageNet-pretrained: 12 blocks, 3 heads, $d = 192$, 5.54 M parameters.

50 epochs | AdamW 5×10^{-4} | cosine decay after 5 warmup epochs | label smoothing 0.1 | AMP | batch 64.

2.83 GPU-hours in total — 14.1 minutes per run, executed unattended.

Isolation

Each run writes its own `outputs/` and `checkpoints/` directory. Three seeds of one arm can never overwrite each other's weights — a failure mode that would have quietly corrupted the whole comparison.

Results you can re-derive, not just read

A run counts as complete only if it emits all four artifacts: `metrics.json`, `best_model.pth`, `last_model.pth`, `train_history.csv`. A failure is logged and the sweep continues; the aggregation step exits non-zero while any of the twelve is missing.

Every table and figure in the paper is written **by script** from one artifact, `outputs/sweep_summary.json`.

Two guards we added after being bitten

The table generator recomputes its statistics independently and **raises** if they disagree with the aggregated summary — and it **refuses to run at all** on an incomplete sweep, rather than emitting a table full of zeros.

Not one number in this deck or in the paper was typed by hand.

05

What the data says

Rufet | Results engine & statistical analysis

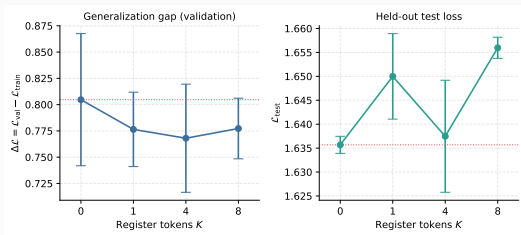
No register configuration improves held-out accuracy

Configuration	Test Top-1 (%)	Test Loss	Val Loss	Gap $\Delta\mathcal{L}$
Baseline ($K = 0$)	75.19 $\pm$ 0.24	1.6357 $\pm$ 0.0018	1.6829 $\pm$ 0.0609	0.8047 $\pm$ 0.0628
Registers ($K = 1$)	74.66 $\pm$ 0.42	1.6500 $\pm$ 0.0089	1.6545 $\pm$ 0.0380	0.7764 $\pm$ 0.0354
Registers ($K = 4$)	75.06 $\pm$ 0.33	1.6375 $\pm$ 0.0117	1.6492 $\pm$ 0.0536	0.7680 $\pm$ 0.0514
Registers ($K = 8$)	74.67 $\pm$ 0.12	1.6560 $\pm$ 0.0022	1.6602 $\pm$ 0.0268	0.7773 $\pm$ 0.0288

The winners split down the middle: the **control** takes both held-out columns, $K = 4$ takes both validation-derived ones.

Paired per seed, $K = 1$ and $K = 8$ lose accuracy in **3 of 3** seeds. $K = 4$ wins in 1 of 3 — $\bar{\delta} = -0.13$ pp, $p = 0.466$.

The finding: a regularizer that only exists on one split



$K = 4$ against the control, paired over seeds:

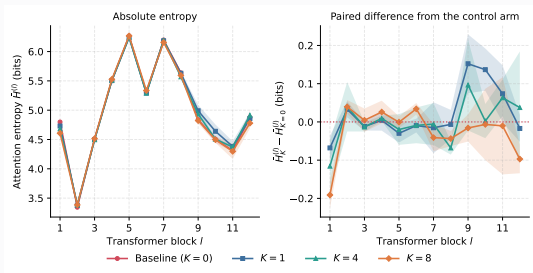
	$\bar{\delta}$	Dir.	p
Validation loss	-0.0337	3/3	0.024
Generalization gap	-0.0367	3/3	0.051
Test loss	+0.0018	2/3	0.823

Training loss across all four arms: **0.8782**, **0.8780**, **0.8812**, **0.8829** — a spread of 0.005.

A regularizer trades training fit for held-out fit. **Nothing was traded.**

Validation spread is **2.17 pp** against **0.24 pp** on test — and it also picks the checkpoint.

H2 supported at $K = 8$; H3 not supported



Left: the four arms lie on top of one another. Right: the paired difference, the only view in which effects this small are legible.

H2 — capacity dilution. $K = 8$ is the study's only consistent held-out effect: test loss $+0.0203$, **3 of 3** seeds, $p = 0.014$. Its final-block entropy is also lowest, falling in 3 of 3.

H3 — entropy collapse. Averaged over twelve blocks the paired differences are $+0.020$, $+0.001$, -0.025 bits. None significant. The patch-norm outlier rate goes $1.227 \rightarrow 1.254\%$ as K grows — it *rises* slightly, the opposite of the prediction.

The artifacts are genuinely there — 1.23% is about nine times the Gaussian 3σ rate. Registers simply do not absorb them here.

Conclusion

What we found

- No register count beats the register-free control on held-out Top-1 ($75.19 \pm 0.24\%$).
- The regularization signature appears on validation and **does not transfer**. Training loss never moved.
- $K = 8$ measurably **costs** held-out loss — 3 of 3 seeds, $p = 0.014$.

The transferable lesson

A generalization-gap argument built on the same split that selects the checkpoint is not evidence about generalization. We nearly published the opposite conclusion from the same twelve runs.

How to read it

A **scope result**, not a refutation. Registers were a pretraining-time intervention at foundation scale; we bolted them onto a pretrained backbone and gave them 50 epochs over 9,000 images. The entropy profile of every register arm is still the *pretrained* profile — the routing never reorganised.

Limits we accept: $n = 3$, one architecture, one budget, diagnostics from a single 64-image batch.

Thank you

Code, sweep runner, figure and table generators:

github.com/shahin1717/vit-research